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A comparative study of the influence of the time-window strategy in online order batching problems --Manuscript Draft--

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Corresponding Author:	Eduardo G. Pardo, Ph.D. Universidad Rey Juan Carlos Móstoles, SPAIN
First Author:	Sergio Gil-Borrás
Order of Authors:	Sergio Gil-Borrás Kenneth Sörensen Ernesto Jiménez Eduardo G. Pardo, Ph.D.
Abstract:	In Online Order Batching Problems (OOBP), orders arrive during a time horizon, and a decision needs to be made on whether to start picking (reducing idle time of the pickers) or to wait for more orders (allowing for more efficient batches configuration). This problem is known as the determination of the Time Window, and it has received far less attention than other decision problems in this context, such as batching and routing. In this paper, we demonstrate that this lack of attention is unwarranted, and that the method used to determine the time window can have a significant impact on the overall performance of the warehouse. To that aim, first, we show that the time-window strategy is independent of the methods used to solve the other subproblems of the OOBP such as batching and routing. Then, we show the behavior of different time-window methods, when varying the number of pickers and the number of orders to be collected. Furthermore, two new time-window methods are proposed in this work that improve in various scenarios the methods previously presented in the state of the art. All conclusions have been experimentally validated over reference instances previously published in the context of the OOBP.
Suggested Reviewers:	Marc Sevaux Université de Bretagne Sud: Universite Bretagne Sud marc.sevaux@univ-ubs.fr Expert on heuristics, metaheuristics and logistics Antonio Alonso-Ayuso Universidad Rey Juan Carlos antonio.alonso@urjc.es Expert on heuristics, metaheuristics. Previous publications on Online Order Batching Problems Abraham Duarte Universidad Rey Juan Carlos abraham.duarte@urjc.es Expert on heuristics, metaheuristics. Previous publications on Online Order Batching Problems



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Eduardo G. Pardo
Universidad Rey Juan Carlos
Department of Computer Science
C/Tulipán S/N, Móstoles
28933, Madrid (Spain)

Professor Yasser Dessouky
Editor-in-Chief, Computers & Industrial Engineering
San Jose State University
Charles W Davidson College of Engineering,
San Jose, California, United States of America

Dear Prof. Dessouky,

I have just submitted the novel research paper titled "**A comparative study of the influence of the time-window strategy in online order batching problems**" that I co-authored with **Sergio Gil-Borrás, Ernesto Jiménez-Merino, and Kenneth Sörensen**, to be considered for publication in **Computers & Industrial Engineering**.

This paper studies the influence of the time-window strategy in the family of Online Order Batching Problems. We have deeply reviewed the state of the art of the strategy, and we have experimentally compared the most outstanding proposals in several contexts. Furthermore, we have proposed two new time-window algorithms. Our findings will help future researchers in the field to improve their algorithms by determining the adequate strategy.

Based in our experience, we truly believe that this is a very nice contribution for the readers of **Computers & Industrial Engineering**.

Yours sincerely,

Eduardo G. Pardo

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Sergio Gil-Borrás

sergio.gil@upm.es

Dept. Sistemas Informáticos. Universidad Politécnica de Madrid

Eduardo G. Pardo

eduardo.pardo@urjc.es

Dept. Computer Science. Universidad Rey Juan Carlos

Ernesto Jiménez

ernesto.jimenez.merino@upm.es

Dept. Sistemas Informáticos. Universidad Politécnica de Madrid

Kenneth Sörensen

kenneth.sorensen@uantwerpen.be

Dept. of Engineering Management. University of Antwerp

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A comparative study of the influence of the time-window strategy in online order batching problems

Highlights

- Concept of time-window strategy in the context of the Online Order Batching Problem
- Literature review of the time-window strategies for Online Order Batching Problem
- Experimental comparison of the best time-window methods in the state of the art
- New algorithmic proposals for the time-window strategy
- An empirical study of several objective functions

A comparative study of the influence of the time-window strategy in online order batching problems

Abstract

In Online Order Batching Problems (OOBP), orders arrive during a time horizon, and a decision needs to be made on whether to start picking (reducing idle time of the pickers) or to wait for more orders (allowing for more efficient batches configuration). This problem is known as the determination of the Time Window, and it has received far less attention than other decision problems in this context, such as batching and routing. In this paper, we demonstrate that this lack of attention is unwarranted, and that the method used to determine the time window can have a significant impact on the overall performance of the warehouse. To that aim, first, we show that the time-window strategy is independent of the methods used to solve the other subproblems of the OOBP such as batching and routing. Then, we show the behavior of different time-window methods, when varying the number of pickers and the number of orders to be collected. Furthermore, two new time-window methods are proposed in this work that improve in various scenarios the methods previously presented in the state of the art. All conclusions have been experimentally validated over reference instances previously published in the context of the OOBP.

Keywords: Online Order Batching Problem, Fixed time-window, Variable time-window, Warehousing, Supply chain

1. Introduction

Logistics in a warehouse encompass a large number of activities. Among them, the collection of orders is one of the most important, due to the high cost associated with this operation in comparison to the rest of the processes. Specifically, it is estimated that the operational cost of order picking can represent up to 60% of the total cost of operations within a warehouse (Drury, 1988; Coyle et al., 1996). This paper focuses on warehouse order picking systems, which follow the batch order picking policy. In this type of system, there are numerous factors that significantly influence the performance of the collection process, such as the shape of the warehouse or the number of aisles. Petersen (1997) published: *“There are four essential factors that greatly influence the performance and efficiency of the order picking operation. They are: (a) layout of the warehouse, (b) the routing and sorting policy, (c) the storage strategy, and (d) the batching method”*. Although we fully agree with the observations made in Petersen (1997), our article focuses on another additional factor, named Time Window (TW) also known as waiting time in this context, not considered by Petersen, which also has a significant influence on the picking time and the completion time. This factor is studied as an additional task that needs to be determined in the context of Order Batching Problems (i.e., those that follow a batch collection policy). Determining the waiting time is especially relevant when the arrival of orders is dynamic (i.e., online), although it could make sense also in multi-picker contexts, when all orders are known before starting the pickup (i.e., offline) but there are potential deadlock situations, such as several pickers trying to access simultaneously to a picking position, to a corridor, or just

26 to the depot. The Online Order Batching Problem (OOBP) is a dynamic
 27 optimization problem where orders continually arrive at the system over the
 28 processing time. In this way, not all the orders to be processed nor their
 29 arrival times are known beforehand. In this context, the batch collection
 30 process can be carried out by a single picker or by several pickers. To solve
 31 the OOBP, it is necessary to tackle different tasks/subproblems. First, the
 32 orders that reach the system are grouped into batches for later collection.
 33 This process of grouping orders into batches is known as *batching*. Once the
 34 batches have been generated, a decision must be made as to whether it is
 35 more convenient to go out and collect any of the batches or wait for new
 36 orders to arrive. This process is known as *determining the time window*. Al-
 37 though at first glance it may seem unnatural (even inefficient) to wait, new
 38 orders may arrive during the time window, which potentially could improve
 39 the distribution of orders in the batches already created. The time window
 40 can be a fixed amount of time or a variable one (i.e., a different waiting time
 41 for each picker at each new route). Furthermore, it is necessary to decide
 42 which batch, among all generated ones, is going to be collected next. This
 43 process is known as *selecting*. Finally, once the batch and the instant for
 44 starting the route are decided, it is necessary to determine the route that
 45 the picker should follow through the warehouse to perform the picking. This
 46 process is known as *routing*. Sometimes, there is an additional process con-
 47 sisting in *sorting* the products collected in a picking route, but this step is
 48 not relevant to our study.

49 This work is motivated by the lack of previous works which specifically
 50 study the task known as determining the time window, in the context of the

OOBP. Particularly, this work can be considered as an extension of the work presented in Gil-Borrás et al. (2020a). However, although there are no studies that compare different algorithms for determining the time window as well as their impact on the picking task, there are works that mention the use of a TW strategy as part of their algorithms for solving different variants of the Online Order Batching Problem. Therefore, this article focuses on studying, in a detailed way, the influence of the time window in the context of the OOBP, depending on different factors such as: batching, routing, objective function, or congestion in the arrival of orders.

The rest of the article is organized as follows. In Section 2, we present the state of the art related to time window. Next, in Section 3, we describe the methods to determine the time window found in the state of the art, and the two new methods proposed in this article. In Section 4, we describe the comparative study carried out. In this section, we also review the algorithms for the rest of the tasks which need to be handled in the context of the OOBP. In Section 5, we present the experiments of this paper. First, we describe the different sets of instances used in this section. Specifically, the best performing methods are compared over five different sets of warehouses previously introduced in the literature. The experimentation carried out compares several methods for determining the time window in combination with different batching and routing algorithms. For each compared method, we perform some preliminary experiments that illustrate the most important strategies. Then, we perform a final experiment to demonstrate the importance of determining the appropriate time window, and its independence from other factors. Finally, in Section 6, some conclusions and

76 pointers for future research are presented.

77 **2. State of the art**

78 The time-window concept emerges in the context of the development of
79 batching algorithms, as an additional strategy. The first study dates from
80 1983 (Quinn, 1983) where the term time-window batching is used as a strat-
81 egy to reduce the picking time. Il-Choe and Sharp (1991) presented several
82 results for its general application in improving the efficiency of a picking sys-
83 tem. Tang and Chew (1997) and Chew and Tang (1999), performed various
84 mathematical analyses related to the use of time-window batching as a batch
85 generation method. Specifically, the upper and lower limits are calculated for
86 the problem, as well as the estimation of the variance and the mean of sev-
87 eral objective functions, such as travel time, service time, and turnover time.
88 Yu and De Koster (2009) expanded previous works by presenting a model
89 based on queuing theory and introducing the calculation of the concept of
90 “Expected waiting time to form a new batch” ($E[W_j]$) applied to this type
91 of problem. The same year, Van Nieuwenhuyse and De Koster (2009) pub-
92 lished an article that estimates the average processing time of an order, using
93 Variable Time-Window Batching (VTWB) or Fixed Time-Window Batching
94 (FTWB) as a batching algorithm. In the same study, the impact of two
95 picking policies (pick-and-sort and sort-while-pick) is also compared for the
96 same Time-Window Batching scenario. This article also develops the con-
97 cept of $E[W_j]$. Later, in 2012, a new method was introduced in Bukchin
98 et al. (2012) for the first time to accurately calculate the departure time of
99 each picker, assuming that all arrival times of orders are known. Based on

100 this information, an approximate model is proposed to determine the wait-
 101 ing strategy for future arrivals of orders. The previous article uses Markov
 102 Decision Processes and they are compared with a couple of naive heuristics.
 103 This work studies the minimization of delays in the delivery of orders, as
 104 well as the costs associated with the overtime of the pickers. The same year,
 105 Henn (2012) presented a metaheuristic algorithm (Iterated Local Search) for
 106 batching, together with a heuristic for calculating the time window.

107 Xu et al. (2014) presented new results using the $E[W_j]$ for the VTWB
 108 calculation, together with the FCFS algorithm for the batching process. In
 109 this paper, authors calculate the optimal size of the batches in this context.
 110 Subsequently, Zhang et al. (2016) first and Zhang et al. (2017) later, intro-
 111 duced a rule-based hybrid heuristic, which linked the computation of the time
 112 window together with the seed-type algorithms for batching. The same year,
 113 Giannikas et al. (2017) proposed three variants of a so-called intervention-
 114 ist strategy, consisting in new Variable Time-Window policies based on the
 115 number of orders that arrive to the system. Duda and Stawowy (2019), used
 116 a Variable Neighborhood Search (VNS) to study the Joint Batch Sequenc-
 117 ing and Picker Routing Problem with time windows. In this work, batches
 118 are formed beforehand and known in advance. Specifically, they divide the
 119 time window into fixed periods of 30 minutes and propose four new neigh-
 120 borhoods to solve the problem in a comprehensive way. Later, Gil-Borrás
 121 et al. (2020a) performed a simple comparison between Fixed Time-Window
 122 and Variable Time-Window methods. Finally, the same year, Leung et al.
 123 (2020) presented the Intelligent B2B order handling system (IOHS), solv-
 124 ing the Integrated Online Pick-to-sort Order Batching using the Fixed and

125 Variable Time-Window Batching strategies.

126 3. Time-window methods

127 According to the literature, a possible classification of time-window meth-
128 ods would divide them into Fixed Time Window and Variable Time Window.
129 In this work, we use these categories to classify the descriptions of the most
130 relevant time-window methods present in the state of the art. Additionally,
131 some of those methods are selected to be empirically compared in Section 5.

132 3.1. Fixed Time-Window methods in the literature

133 Fixed Time-Window (FTW) methods are characterized by determin-
134 ing a fixed waiting time that can depend on different factors of the sys-
135 tem/instance. Specifically, the strategy that determines the time window
136 could set a single fixed time for all instances of the problem. Alternatively,
137 a FTW strategy could also set a different fixed time for each instance, but
138 depending on a parameter of the instance. The most representative Fixed
139 Time-Window methods in the literature are described below:

- 140 • **No-wait method (FTW_NW):** this is the simplest FTW method.
141 It consists of starting the picking of the next batch, as soon as there
142 is a picker available and a batch waiting to be picked. This strategy is
143 one of the most used in the literature. In fact, if not stated otherwise,
144 it is taken as the default strategy. Some examples of works that use
145 this strategy are Gil-Borrás et al. (2020a,b, 2021).
- 146 • **Methods with the same fixed time for all instances (FTW_CT):**
147 it consists of establishing, following any criteria, a fixed time that a

picker must wait, each time the picker is available, regardless of the specific instance in which the picker is working. This strategy was explored in Gil-Borrás et al. (2020a), where different time-window methods are compared. Specifically, the authors in this paper demonstrate how, in general, this type of method returns worse solutions than methods which use a Variable Time-Window strategy.

- **Methods with a calculated fixed time for each instance (FTW_ZH):** these methods can be considered as a particularization of the FTW_CT. Specifically, they consist of setting a fixed time based on a calculation per instance. An example of the use of this type of methods was described in Zhang et al. (2017) although it is based on the studies of Xu et al. (2014) and Van Nieuwenhuyse and De Koster (2009). In this work, authors determined the expected arrival time of the next k orders, required to complete a batch, on the basis of the arrival information of previous orders.

We have selected FTW_NW and FTW_ZH methods to be experimentally evaluated in Section 5 since according to the previous comparison (Gil-Borrás et al., 2020a) FTW_CT methods performed worse.

3.2. Variable Time-Window methods in the literature

Variable Time-Window (VTW) methods are characterized by having a variable waiting time, which can be different at each picking moment for the same instance depending on different factors. The most representative methods in the literature are described below:

- 171 • **Methods based on a minimum number of orders in queue**
172 **(VTW_QO)**: this strategy consists of starting the collection of or-
173 ders in a batch when there is a minimum number of pending orders
174 to be collected in the system. This method was recently used in Gil-
175 Borrás et al. (2020a), where the authors explored the performance of
176 this strategy for different number of queued orders (4, 8, and 16).

- 177 • **Methods based on a minimum number of batches (VTW_BA)**:
178 this method sets a different time window depending on the completion
179 of a minimum number of batches previously defined, prior to starting
180 picking. The simplest version of this strategy consists of waiting until
181 the first batch is full. Particularly, when a new batch is configured, it
182 is assumed that the previous one does not have the capacity to accom-
183 modate new orders, so at that time the picker can start collecting the
184 first batch, and so on.

- 185 • **Reactive methods according to the collection conditions (VTW-**
186 **HE)**: this method determines the time window on the basis of the
187 specific system conditions. For example, Henn (2012) used a variant
188 of this method, where the time window was determined depending on
189 the arrival and service time of the previous order.

- 190 • **Methods based on data-built models (VTW_MM)**: in addition
191 to the previously introduced heuristic methods, there is a family of
192 methods based on machine learning models. Specifically, in Bukchin
193 et al. (2012) the authors propose a method that uses Markov decision
194 processes to calculate a subsequent model that determines the optimal

195 moment in which a picker should have started the picking of each batch.
196 Based on the model built with previous data, it establishes a decision-
197 making system for future orders. However, the construction of the
198 exact model presents the difficulty of being a computationally costly
199 task.

200 We have selected the VTW_BA and VTW_HE methods to be experimen-
201 tally evaluated in Section 5, since according to a previous comparison (Gil-
202 Borrás et al., 2020a) perform better than VTW_QO. Additionally, VTW_MM
203 is impractical for real-size instances as the ones used in this paper.

204 3.3. *New time-window methods*

205 In addition to the previous methods identified in the state of the art,
206 in this work we propose two new VTW methods for determining the time
207 window. These methods have been tested on different empirical scenarios
208 and improve the methods in the state of the art in several of them. The
209 experiments are reported in Section 5.

- 210 • **Method based on the available capacity (VTW_M1):** this method
211 considers that a picker should wait while there is a batch under con-
212 struction still incomplete, together with a high estimated probability
213 that the next order will fit in the available capacity. Since the method
214 is based on estimations, in the event that the next order that arrives
215 in the system exceeds the available capacity (i.e., generating a new
216 batch), the picker will start collecting the previous batch. On the
217 other hand, while the capacity is not exceeded after a new arrival, the
218 method constructs a probability distribution based on the size of the

orders previously arrived to the system. Based on this distribution, it calculates the probability of the next order of having a smaller size than or equal to the available capacity in the current batch. Then, based on a threshold, the method decides whether to start picking or to wait. The probability threshold is a search parameter and should be experimentally adjusted.

- **Method based on the available capacity with rules (VTW_M2):** this method could be considered an extension of VTW_M1, but adding two new criteria to start picking: 1) the time calculated for the picking task of the orders currently in the batch under construction is smaller than the time needed until the estimated instant of arrival of the next order; 2) the average time required to pick the items assigned to the batch under construction is at least 10% smaller than the average picking time of items previously collected.

The VTW_M1 and VTW_M2 methods have been included in the experimental evaluation in Section 5.

4. Description of the comparative study

To evaluate the time-window algorithms exposed in the Section 3, it is necessary to define the strategies that will be used for the rest of the activities in the context of the OOBP. Also, it is necessary to determine the objective functions to be optimized.

For variants with a single picker, it is necessary to define the batching, selecting, and routing strategies. Moreover, if there are multiple pickers, it is necessary to determine an additional task known as *assigning*.

243 In this work, we study different batching and routing strategies, which
244 are described in Section 4.1 and Section 4.2, respectively. On the other hand,
245 the selecting method used consists of selecting the batch that contains the
246 largest number of items. In the event that there is more than one batch
247 with the maximum number of items, we select the batch with the shortest
248 picking route. Additionally, the assigning method, which decides which batch
249 is collected by which picker, assigns the next batch to be collected to the
250 first available picker. Finally, we study the minimization of three different
251 objective functions: picking time, completion time, and maximum turnover
252 time, which are described in Section 4.3.

253 *4.1. Batching strategies*

254 The batching task consists of grouping a set of orders in a batch which
255 will be collected together in a single picking route. We consider that orders
256 can not be split in different batches. There is a great variety of algorithms
257 published in the literature for this task. In this work, we compare three
258 different batching strategies: First Come First Served (FCFS), Iterated Local
259 Search (ILS), and General Variable Neighborhood Search (GVNS). The
260 purpose of exploring different batching strategies is to determine if the time-
261 window method selected depends on the batching algorithm used. Next, we
262 review the batching strategies considered.

263 *4.1.1. First Come First Served*

264 FCFS is a simple heuristic algorithm that sorts the orders on the basis
265 of the arrival time of each order to the system. Then, batches are generated
266 starting from an empty one and adding orders to the solution one by one

267 following the previous predefined sequence, starting from the oldest order.
268 Once an order does not fit in the current batch, that batch is considered com-
269 plete and a new batch is generated. The process is repeated until all orders
270 have been assigned. This simple algorithm is widely used in the literature,
271 and it is usually considered as a baseline method for comparison.

272 4.1.2. *Iterated Local Search*

273 ILS metaheuristic was devised in Lourenço et al. (2003) as a method
274 based on local search to perform an efficient exploration of the solution space.
275 Specifically, it consists of alternating an improvement phase with a perturba-
276 tion phase. The former is devoted to reach a local minimum, while the latter
277 is used for diversifying the search. ILS algorithm was used in the context of
278 the OBP in Henn et al. (2010). In this work, the improvement phase uses
279 two neighborhoods: the first one is based on the exchange operation of two
280 orders located in two different batches (swap); and the second one is based
281 on the insertion of an order in a different batch from the current one (shift).
282 On the other hand, the perturbation phase is based on the exchange of a
283 random number of orders between batches selected at random.

284 4.1.3. *General Variable Neighborhood Search*

285 The VNS methodology was initially proposed in Mladenović and Hansen
286 (1997) as a search strategy based on the concept of neighborhood change to
287 escape from local optima. Among the multiple VNS variants, Variable Neigh-
288 borhood Descent (VND) is characterized by systematically exploring several
289 neighborhoods using a local search procedure. General Variable Neighbor-
290 hood Search (GVNS) uses VND as an intensification strategy, but adds a

291 perturbation strategy for diversification purposes. In this work, a GVNS al-
 292 gorithm has been used in combination with the VND procedure proposed in
 293 Gil-Borrás et al. (2020b) for the OOBP context. This algorithm uses three
 294 neighborhoods based on the following operations: exchange of two orders
 295 belonging to a different batch (1x1); exchange of two orders belonging to a
 296 batch with an order belonging to a different batch (2x1); and insertion of an
 297 order in a different batch than the current one (1x0). The neighborhoods
 298 and their order have been established as indicated by the authors. The lo-
 299 cal search procedures proposed use a first improvement strategy. Finally,
 300 the perturbation phase consists of performing a random exchange of orders
 301 between orders belonging to different batches.

302 *4.2. Routing strategies*

303 The routing task consists of generating a route that enables the collection
 304 of the selected set of orders in a batch, within the warehouse. This strategy
 305 is also used to determine aspects such as picking time or distance traveled.
 306 In general, there are three families of strategies to generate the route: ex-
 307 act methods, heuristic methods, and metaheuristic methods. This research
 308 focuses on heuristic methods as they are the most popular in either the lit-
 309 erature and real scenarios, because they are easy to implement and easy to
 310 understand by pickers.

311 In this work, three routing heuristic algorithms have been compared: S-
 312 shape, Largest-gap, and Combined for demonstrating that the time-window
 313 strategy is independent of the routing algorithm used. An empirical compar-
 314 ison of these three procedures can be found in Petersen (1997) and Petersen
 315 and Aase (2004).

316 4.2.1. *S-Shape*

317 The S-Shape method is a heuristic strategy where pickers start their
318 routes in the depot that is located in the front cross aisle. Pickers move
319 through the front cross aisle until they reach the leftmost parallel aisle with
320 items to collect in the warehouse. Then, the pickers fully traverse this parallel aisle and any others which contain items to collect. To change from
321 one parallel aisle to another, the picker can use either the front cross aisle or
322 the back cross aisle indistinctly, depending on the route. However, pickers
323 must finish their route in the depot again, so in the event that the number
324 of parallel aisles to traverse is odd, the last aisle will only be traversed until reaching the last item to be collected and then, the pickers turn around
325 and come back to the front cross aisle to return to the depot and finish the
326 picking route. We refer the reader to Hall (1993) and Petersen (1995) for a
327 further description of this strategy.

330 4.2.2. *Largest Gap*

331 The Largest-Gap method is a heuristic strategy based on the concept
332 of gap. The gap is defined as the distance between two items to be picked
333 consecutively and located in the same parallel aisle, or also the distance
334 between a cross aisle and the first item to be picked in a parallel aisle. Again,
335 the picking route begins and ends in the depot, which is located in the front
336 cross aisle. Notice that the entrance to the parallel aisles can be made from
337 any of the two crossing aisles (either the back or the front cross aisle) and
338 pickers only enter the aisles that have items to collect. The Largest-Gap
339 strategy consists of avoiding going through the part of the aisle with the
340 largest gap. To that aim, when the pickers arrive to the position where the

largest gap starts, they turn around in the parallel aisle and go back to the initial cross aisle. This procedure is performed except for the first and last parallel aisles with items to collect, which are fully traversed. Therefore, pickers might enter in the same aisle twice: once from the front cross aisle and once from the back. We refer the reader to Hall (1993) and Petersen (1995) for a further description of this strategy.

4.2.3. *Combined*

Finally, the Combined heuristic strategy consists of the combination of the two previous heuristics: S-Shape and Largest Gap. Specifically, the Combined method uses the lowest cost technique (S-Shape or Largest Gap) in each aisle with items to collect. As in the previously studied methods, pickers begin their route at the depot, which is located in the front cross aisle. Pickers traverse the front cross aisle until they reach the leftmost parallel aisle with items to collect, deciding the best strategy for performing picking in that and subsequent aisles. Notice that the restrictions associated with either Largest Gap or S-Shape strategies must be satisfied. Once all items have been collected, the pickers return to the depot. We refer the reader to De Koster and Van Der Poort (1998) and Menéndez et al. (2017b) for a further description of this strategy.

4.3. *Objective functions*

In this work, three widely used objective functions for different variants of the OBP are compared: the minimization of the picking time; the minimization of the completion time; and the minimization of the maximum turnover time.

365 The picking time is the time that pickers need to perform the picking task
366 of all orders, once the batches are conformed. The completion time is similar
367 to the picking time, but also includes the waiting time for the arrival of new
368 orders. Finally, the maximum turnover time is the maximum time that an
369 order remains in the system since it arrives until it is served.

370 5. Experiments

371 This section presents different experiments to evaluate the time-window
372 methods previously detailed in the context of the Online Order Batching
373 Problem. Also, we study the relationship between the time window strategies
374 and other strategies such as routing or batching algorithms, as well as the
375 influence of either the set instances or the optimized objective function on
376 the performance of the methods.

377 All methods used in the experimentation, including those of the state of
378 the art, were coded in Java 8 and run on an Intel (R) Core (TM) 2 Quad
379 CPU Q6600 2.4 Ghz computer, with 4 GB DDR2 RAM memory and Ubuntu
380 18.04.1 64 bit LTS operating system.

381 In Section 5.1, we present the sets of instances used in the comparison.
382 In Section 5.2, we perform a set of preliminary experiments to adjust the
383 values of the different compared time-window algorithms. Finally, in Section
384 5.3 we present the final experiments. In those experiments, we compare
385 the behavior of the routing and batching algorithms when combined with
386 different time-window strategies.

387 5.1. *Instances*

388 Two sets of instances widely used in the state of the art of different
 389 variants of the OBP have been selected for this article. The objective of
 390 using two sets of instances is to evaluate if there is any dependence of the
 391 time-window strategies on the set of instances used.

392 The first data set (Dataset #1) is composed of 80 instances corresponding
 393 to 4 different warehouses. It was originally proposed in Albareda-Sambola
 394 et al. (2009) and it has been used in many related works (Gil-Borrás et al.,
 395 2020b; Menéndez et al., 2017b; Menéndez et al., 2015).

396 The second data set (Dataset #2) is composed of 64 instances correspond-
 397 ing to a single warehouse. It was originally proposed in Henn (2012) and it
 398 has also been used in many related papers (Aerts et al., 2021; Alipour et al.,
 399 2020; Gil-Borrás et al., 2020b; Koch and Wäscher, 2016; Menéndez et al.,
 400 2017a,b; Pérez-Rodríguez et al., 2015).

401 Instances in both data sets correspond to single-block rectangular ware-
 402 houses with two cross aisles, one at the front and one at the back. The cross
 403 aisles are linked by a number of parallel aisles containing shelves with picking
 404 positions at both sides of the aisle. The depot is the place in the warehouse
 405 where pickers start and end the picking routes and where the products are
 406 deposited once they have been collected. We consider that there is a single
 407 depot in the warehouse that is located in the center or in the left corner of the
 408 front cross aisle, depending on the instance. These data sets can be down-
 409 loaded at <http://grafo.etsii.urjc.es/opticom/oobp/>. In Table 1, we compile
 410 the main characteristics of the instances used in our experiments.

411 Additionally, since these experiments occur in a dynamic/online con-

	Dataset #1 (Albareda-Sambola et al. (2009))				Dataset #2 (Henn (2012))			
	W1	W2	W3	W4	W5			
Storage policy	Random / ABC				Random / ABC			
Depot position	Center / Left corner				Center			
Order size	U(1,7)	U(2,10)	U(5,25)	U(1,36)	U(5,25)			
Item weight	1	1	1	U(1,3)	1			
Batch capacity (weight)	12	24	150	80	30 / 45 / 60 / 75			
Number of parallel aisles	4	10	25	12	10			
Number of items per aisle	2x30	2x20	2x25	2x16	2x45			
Number of items	240	400	1250	384	900			
Parallel aisle length	50 m	10 m	50 m	80 m	45 m			
Parallel aisle width	4.3 m	2.4 m	5 m	15 m	5 m			
Number of instances	20	20	20	20	64			
Travel speed (m/min.)	48	48	48	48	48			
Extraction speed (items/min)	6	6	6	6	6			
Batch setup time	3 min	3 min	3 min	3 min	3 min			

Table 1: Characteristics of the warehouses and the work parameters.

	#Customer orders							
	40	50	60	80	100	150	200	250
4 Hours	0.167	0.208	0.250	0.334	0.417	0.625	0.834	1.042

Table 2: λ values for each number of orders.

412 text, where orders arrive dynamically (i.e., once the picking task has already
 413 started) it is also necessary to model the arrival of the orders to the system.
 414 In this case, we have studied a scenario where the arrival of orders follows
 415 an exponential distribution derived from a Poisson process (Poisson, 1837)
 416 characterized by a time horizon of 4 hours ($t = 4$). The number of events
 417 (the arrival of orders in this context) that occur in a time interval t is a
 418 random variable $X(t)$ with mean $E[X(t)] = \lambda * t$. The λ value is selected
 419 depending on the number of orders considered for each instance. In Table 2
 420 we report the values of λ for each considered case.

421 5.2. Preliminary experiments

422 This section is devoted to tuning the parameters of the compared time-
423 window algorithms: FTW_ZH (Zhang et al., 2017), VTW_HE (Henn, 2012),
424 VTW_M1, and VTW_M2 (proposed in this paper). To carry out these ad-
425 justments, a diverse subset of 24 instances has been selected from the orig-
426 inal sets of instances previously described. It is worth mentioning that, in
427 the case of VTW_BA, three different configurations (2, 3, and 4 generated
428 batches) are used in the final experiments, denoted as: VTW_B1, VTW_B2,
429 and VTW_B3, respectively.

430 5.2.1. Selection of parameter β in FTW_ZH

431 The time-window method proposed in Zhang et al. (2017) uses a β pa-
432 rameter in the formula that determines the fixed time interval of the time
433 window. Specifically, β is a coefficient of the average picking time of q dif-
434 ferent items. In this experiment, different values of β (0.5, 1, and 1.5) and
435 their influence are compared.

436 In Table 3 we compare, for each value of β , the average value of the pick-
437 ing time measured in seconds, Avg.(s), the deviation to the best solution of
438 the experiment, Dev.(%), and the number of best solutions found, #Best.
439 Similarly, the same metrics are reported for the completion time. Observ-
440 ing the results, the configuration which obtained a larger number of best
441 solutions either in picking time or in completion time was $\beta = 1.5$ with a
442 good performance in the other two indicators. Therefore, we have used this
443 configuration for the final experiments.

		β Value		
		0.5	1.0	1.5
Picking time	Avg.(s)	41291	41300	41293
	Dev.(%)	0.17%	0.31%	0.18%
	#Best	10	9	13
Completion time	Avg.(s)	41955	41967	41950
	Dev.(%)	0.20%	0.36%	0.19%
	#Best	7	7	10

Table 3: Study of different values for the β parameter in FTW_ZH method for picking time and completion time.

444 5.2.2. Selection of parameter α in method VTW_HE

445 The time-window method proposed in Henn (2012) uses a parameter α to
446 determine the time window before starting the picking of the next batch un-
447 der construction. Specifically, α is a coefficient that weights the arrival time
448 of the last order and the service time of the current batch under construction,
449 in the formula for calculating the time window. In this experiment, different
450 values of α (25%, 50%, 75%, and 100%) are compared. The value 0% for α
451 is not evaluated as it would be equivalent to the FTW_NW method that is
452 already included in the final experiments.

453 In Table 4 we show the three previously introduced quality indicators for
454 the considered values of α for both objective functions. In this case, it can be
455 observed that the best value for the parameter α is 50% in the three indicators
456 for the two objective functions studied. For this reason, $\alpha = 50\%$ is selected
457 for the configuration of the method VTW_HE in the final experiments.

458 5.2.3. Study of the capacity threshold in VTW_M1

459 The VTW_M1 method determines the probability that the next order
460 arriving in the system will fit into the available capacity in the batch under

		α Value			
		25%	50%	75%	100%
Picking time	Avg.(s)	40515	40430	40483	40518
	Dev.(%)	0.61%	0.51%	0.57%	0.73%
	#Best	6	12	7	5
Completion time	Avg.(s)	42121	42022	42085	42111
	Dev.(%)	1.06%	0.83%	0.97%	1.02%
	#Best	3	7	3	3

Table 4: Study of different values for the α parameter in FTW_HE method for picking time and completion time.

461 construction. In this experiment, we study different thresholds to start pick-
462 ing if the probability is below that threshold. Specifically, we have evaluated
463 threshold values between 20% and 70% (in steps of 10%). The extreme val-
464 ues have not been evaluated as a threshold of 0% indicates that the next
465 order would not fit in the batch, therefore there is no point in waiting for it.
466 Similarly, a value of 100% would equate to the batch being empty.

467 In Table 5, we report the same quality indicators previously introduced
468 for the different threshold values evaluated, both for picking time and for
469 completion time. In this case, the best value for this parameter in the three
470 proposed indicators, for both objective functions, is 60%. Therefore, this
471 value has been selected for the configuration of VTW_M1 in the final exper-
472 iments.

473 5.2.4. Study of the capacity threshold in VTW_M2

474 The VTW_M2 method determines the probability that the next order
475 arriving to the system will fit into the available capacity (as VTW_M1) but
476 including several additional rules. Specifically, we have evaluated threshold
477 values between 20% and 70% (in steps of 10%). Again, the extreme values
478 have not been evaluated as a threshold of 0% indicates that the next order

		Threshold (%)					
		20%	30%	40%	50%	60%	70%
Picking time	Avg.(s)	40738	40793	40779	40856	40721	40813
	Dev.(%)	0,24%	0,38%	0,36%	0,31%	0,21%	0,33%
	#Best	7	4	4	6	8	5
Completion time	Avg.(s)	41725	41776	41760	41839	41701	41799
	Dev.(%)	0,23%	0,37%	0,34%	0,30%	0,19%	0,34%
	#Best	4	1	5	6	6	2

Table 5: Study of different threshold values in VTW_M1 method for the picking time and completion time.

would not fit in the batch, therefore there is no point in waiting for it. Similarly, a value of 100% would be equivalent to always waiting.

In Table 6, we show the same three quality indicators previously presented. In this case, the best value for this parameter, for both objective functions, is 40% for all the proposed indicators except for the number of best solutions in the completion time case. Therefore, this value has been selected for the configuration of VTW_M2 for the final experiments.

		Threshold (%)					
		20%	30%	40%	50%	60%	70%
Picking time	Avg.(s)	40359	40335	40285	40334	40310	40295
	Dev.(%)	0,43%	0,47%	0,27%	0,36%	0,34%	0,41%
	#Best	6	4	6	6	5	3
Completion time	Avg.(s)	40858	40838	40784	40837	40815	40790
	Dev.(%)	0,41%	0,47%	0,28%	0,35%	0,35%	0,39%
	#Best	4	5	3	5	5	2

Table 6: Study of different threshold values in VTW_M2 method for the picking time and completion time.

5.3. Final experiments

The aim of this work is to evaluate the behavior of the previously selected strategies for determining the time window in different scenarios. To that

489 aim, we have divided our final experimentation in four sections where we
 490 vary: the routing strategy, the batching strategy, or the characteristics of
 491 the instance (number of pickers and congestion). Our objectives are: 1)
 492 study the influence of the time-window algorithm on the performance of the
 493 methods depending on the objective function; 2) identify the best strategy
 494 for determining the time window in each scenario; and 3) determine if there
 495 exists a dependency of the waiting strategy with respect to either the batching
 496 or the routing strategies, or other factors such as the number of pickers or
 497 the congestion in the arrival of orders.

498 Notice that all experiments use the same selection and assignment strate-
 499 gies (see Section 4). Additionally, in those experiments where the number
 500 of pickers in the warehouse is not explicitly indicated, it is considered that
 501 there is only one picker. All experiments in this section have been performed
 502 over the whole data set of instances introduced in 5.1.

503 *5.3.1. Impact of time-window algorithms on the performance of the studied*
 504 *methods for different objective functions, when combined with several*
 505 *routing strategies*

506 In this experiment, we study the picking time and the completion time
 507 objective functions when varying the routing and waiting strategies. Par-
 508 ticularly, we combined a GVNS batching method (see Section 4.1.3) with
 509 three different routing algorithms (S-Shape, Largest-Gap, and Combined) in-
 510 troduced in Section 4.2, and with eight time-window algorithms (VTW_B1,
 511 VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW-
 512 _ZH) introduced in Section 3.

513 In Figure 1, we represent the different algorithmic variants compared

514 in terms of completion time and picking time. Particularly, we report the
515 average among all instances of the minimum completion time or the minimum
516 picking time. It can be observed that the time window method substantially
517 influences the results obtained (either in terms of picking time or in terms of
518 completion time) regardless of the routing method. Also, it is observed that
519 the influence of each time-window strategy on the final result (both in picking
520 time and in completion time) remains constant for each routing method.
521 That is, the best time-window method when using Combined, Largest Gap,
522 or S-Shape routing method is always VTW_B3 in terms of picking time. It is
523 also observed that the best time-window method to minimize the picking time
524 (VTW_B3) is not the best time-window method to minimize the completion
525 time. In this case, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH, behave
526 very similar one to each other and they can be considered the best methods
527 among the compared ones for the completion time.

528 Similarly, in Figure 2, we compare the performance of the previous meth-
529 ods in terms of maximum turnover time, in comparison with the completion
530 time. In this case, it is confirmed that, whatever the routing method is, the
531 use of one or another time-window method substantially influences the result
532 obtained in terms of maximum turnover time. Again, we report the average
533 obtained values among all instances. In this case, similarly to what happens
534 with the completion time, the methods VTW_M1, VTW_M2, FTW_NW, and
535 FTW_ZH perform alike and they can be considered as the best ones among
536 the compared methods, for minimizing the maximum turnover time.

537 The detailed results to elaborate Figure 1 and Figure 2 can be found in
538 Appendix A.

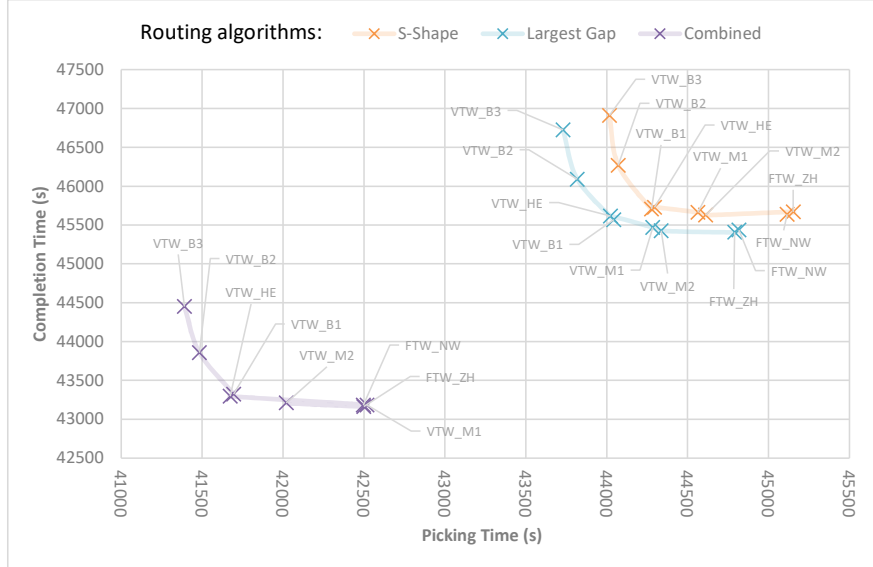


Figure 1: Behavior of the picking time and completion time when combining the GVNS batching algorithm, with different routing strategies (S-Shape, Largest-Gap, and Combined) and different time-window strategies (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH).

539 *5.3.2. Impact of time-window algorithms on the performance of the studied*
540 *methods for different objective functions, when combined with several*
541 *batching strategies*

542 In this experiment, we study the picking time and the completion time
543 objective functions when varying the batching and waiting strategies. Par-
544 ticularly, we combined three different batching methods (GVNS, ILS, and
545 FCFS) described in Section 4.1, with the S-Shape routing strategy (see Sec-
546 tion 4.2.1), and with eight time-window algorithms (VTW_B1, VTW_B2,
547 VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) in-
548 troduced in Section 3.

549 In Figure 3, we represent the different algorithmic variants compared in
550 terms of completion time and picking time. As it can be observed, what-

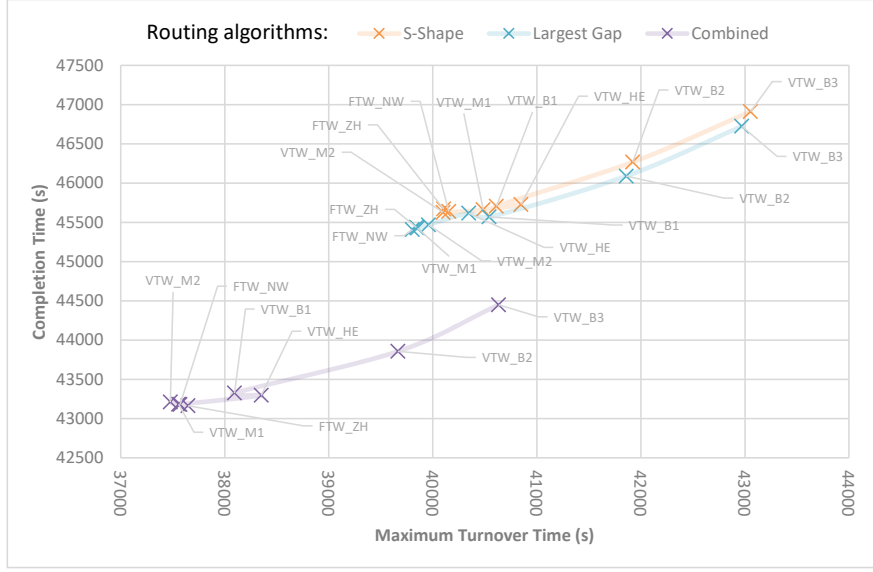


Figure 2: Behavior of the maximum turnover time and completion time when combining the GVNS batching algorithm, with different routing strategies (S-Shape, Largest-Gap, and Combined) and different time-window strategies (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH).

551 ever the batching method is, the use of one or another time-window method
 552 substantially influences the results obtained (either in terms of picking time
 553 or in terms of completion time). Also, it is observed that, regardless of the
 554 chosen batching method, the influence of each time-window strategy on the
 555 final result (both in picking time and in completion time) remains constant.
 556 That is, the best time-window method for picking time when using GVNS
 557 or FCFS is VTW_B3, while VTW_B2 is slightly better than VTW_B3 when
 558 combined with ILS. It is also observed that the best time-window method to
 559 minimize the picking time (VTW_B3) is not the best time-window method
 560 to minimize the completion time. In this case, again, VTW_M1, VTW_M2,
 561 FTW_NW, and FTW_ZH are the best methods and behave similar one to
 562 each other for any of the batching methods compared.

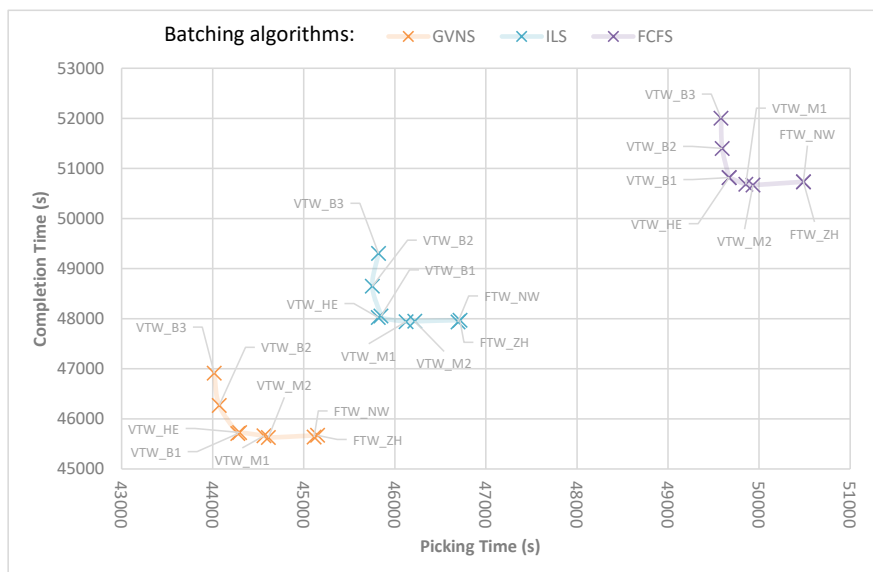


Figure 3: Behavior of the picking time and completion time when combining different batching algorithms (GVNS, ILS, and FCFS), with several time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) using the same routing algorithm (S-Shape).

Similarly, in Figure 4, we compare the performance of the previous meth-
ods in terms of maximum turnover time when compared with with the com-
pletion time. In this case, it is confirmed that, whatever the batching method
is, the use of one or another time-window method substantially influences
the result obtained in terms of maximum turnover time. In this case, sim-
ilar to what happens with the completion time, the methods VTW_M1,
VTW_M2, FTW_NW, and FTW_ZH perform alike and they can be con-
sidered as the best ones among the compared methods, for minimizing the
maximum turnover time, for any of the batching strategies compared.

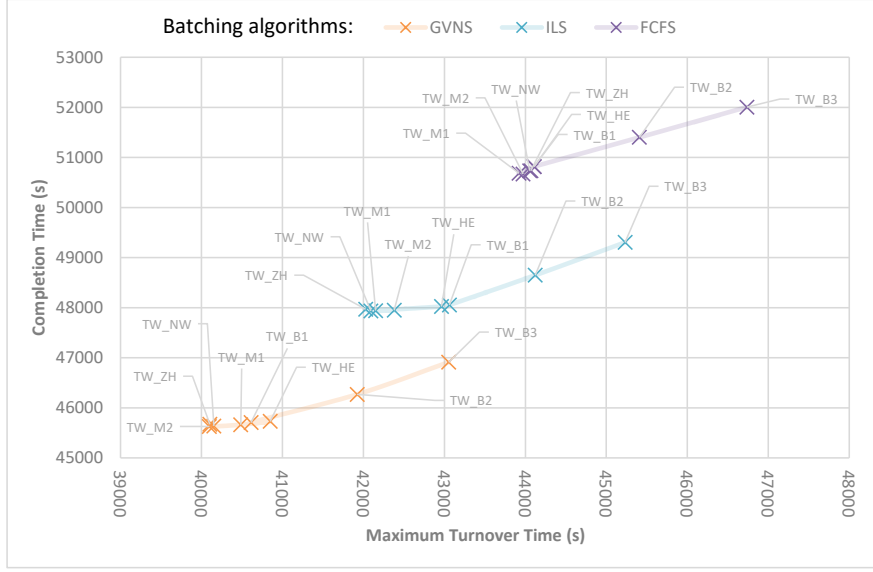


Figure 4: Behavior of the maximum turnover time and completion time when combining different batching algorithms (GVNS, ILS, and FCFS), with several time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) using the same routing algorithm (S-Shape).

572 The detailed results to elaborate Figure 3 and Figure 4 can be found in
573 Appendix B.

574 *5.3.3. Impact of time-window algorithms on the performance of the studied*
575 *methods for different objective functions, when varying the number of*
576 *pickers*

577 In this experiment, we study the picking time and the completion time
578 objective functions, when varying the number of pickers and the waiting
579 strategies. Particularly, we combined the GVNS batching strategy (see Sec-
580 tion 4.1.3), with the S-Shape routing strategy (see Section 4.2.1), and with
581 eight time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE,
582 VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) introduced in Section 3 for

583 scenarios with a different number of pickers (1, 2, 3, 4, and 5). The results
 584 of this experiment are presented in Figure 5.

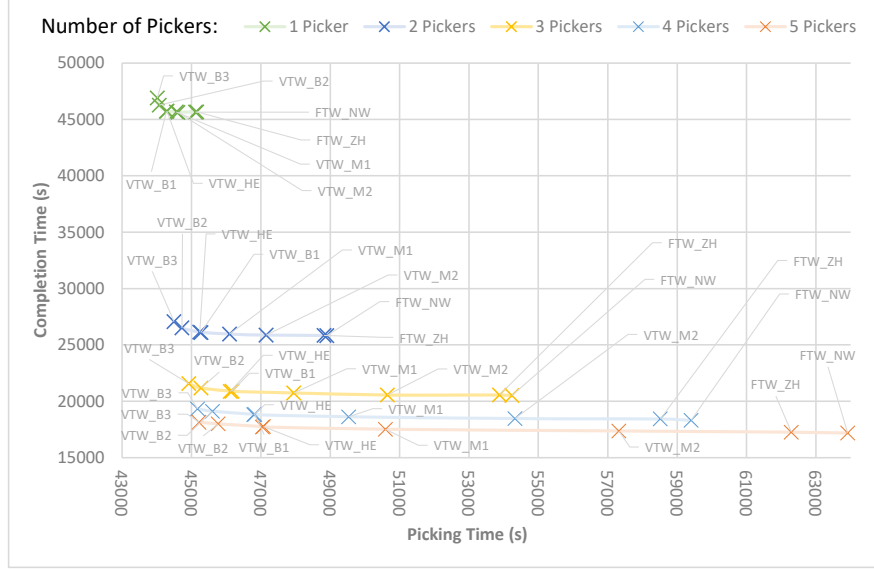


Figure 5: Behavior of the picking time and completion time when increasing the number of pickers (1, 2, 3, 4, and 5 pickers) for different time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH), using a GVNS batching algorithm and a S-Shape routing strategy.

585 As expected, there is an increase in the picking time when increasing the
 586 number of pickers, but also a decrease in the completion time. On the other
 587 hand, it is observed that the method configured with VTW_B3 is the best
 588 in terms of picking time for any scenario (1, 2, 3, 4, and 5 pickers). Simi-
 589 larly, the methods FTW_NW, FTW_ZH, VTW_M1, and VTW_M2 present
 590 a similar behavior in terms of completion time, and can be considered as the
 591 best methods for the minimization of this objective function, in any of the
 592 scenarios studied (1, 2, 3, 4, and 5 pickers).

593 Finally, it is observed that the differences between the best and the worst
 594 methods for the same number of pickers are very small when considering

the completion time, regardless of the time window method. However, in the case of the picking time, the differences increase substantially when the number of pickers increases.

Similarly, in Figure 6, the same methods are compared in terms of maximum turnover time and completion time. In the case of the maximum turnover time, it is generally observed that it decreases as the number of pickers increases. More specifically, observing the algorithm configured with the same time-window strategy but for a different number of pickers, both the maximum turnover time and completion time decrease as the number of pickers increases. Furthermore, when the number of pickers increases, the influence of the time-window strategy is more relevant in terms of maximum turnover time, since the difference between the worst and the best methods also increases. Derived from the experiments carried out, the methods FTW_NW, FTW_ZH, VTW_M1, and VTW_M2 (which behavior is similar in terms of maximum turnover time) can be considered the best methods for the minimization of the maximum turnover time objective function, in any of the scenarios studied (1, 2, 3, 4, and 5 pickers).

The detailed results to elaborate Figure 5 and Figure 6 can be found in Appendix C.

5.3.4. Impact of time-window algorithms on the performance of the studied methods for different objective functions, when varying the congestion in the arrival of orders

In this last experiment, we compare the picking time, the completion time, and the maximum turnover time for different variants of an algorithm configured with the GVNS batching strategy (see Section 4.1.3), the S-Shape

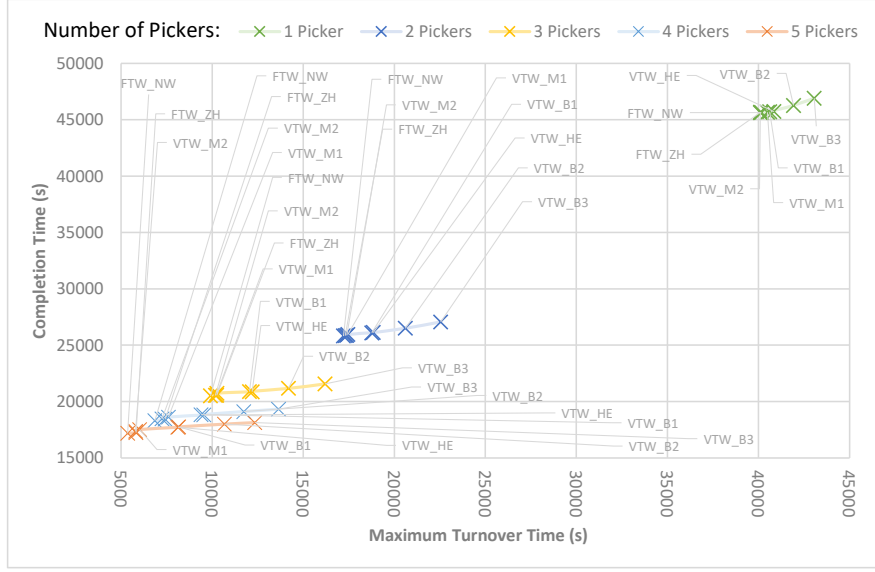


Figure 6: Behavior of the maximum turnover time with respect to the completion time when the number of pickers is increased (1, 2, 3, 4, and 5 pickers), for several time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH), using a GVNS batching algorithm and a S-Shape routing strategy.

620 routing strategy (see Section 4.2.1), and each of the eight compared time-
621 window strategies (see Section 3), for instances with different number of
622 orders received within the same time horizon (i.e., varying the congestion in
623 the system). Specifically, the set of instances extracted from Dataset #2 in
624 Henn (2012) has a size which varies from 40 to 100 orders, while the set of
625 instances extracted from Dataset #1 in Albareda-Sambola et al. (2009) has
626 a size which varies from 50 to 250 orders. The results are presented in Tables
627 7-12 ordered by the objective function (picking time, completion time, and
628 maximum turnover time) and set of instances. Additionally, for each table,
629 the results are organized depending on the number of orders.

		Picking time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
40	Avg (s)	15966	16071	16544	16501	16919	16990	17647	17639
	Dev (%)	0.22%	0.94%	4.22%	3.90%	6.67%	7.25%	12.20%	12.16%
	#Best	10	5	1	0	0	0	0	0
60	Avg (s)	23066	23176	23352	23368	23574	23618	24123	24087
	Dev (%)	0.13%	0.72%	1.45%	1.63%	2.59%	2.77%	5.57%	5.40%
	#Best	12	3	0	0	1	0	0	0
80	Avg (s)	30783	30843	30956	30931	31134	31061	31372	31423
	Dev (%)	0.14%	0.35%	0.72%	0.65%	1.36%	1.19%	2.37%	2.56%
	#Best	8	6	0	1	0	0	1	1
100	Avg (s)	36680	36694	36817	36875	37066	37115	37225	37347
	Dev (%)	0.28%	0.31%	0.64%	0.78%	1.32%	1.44%	1.91%	2.18%
	#Best	7	7	2	0	0	0	0	0

Table 7: Behavior of the picking time when varying the number of orders (congestion), for several time-window strategies (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) using the GVNS batching algorithm and the S-Shape routing strategy on the instances extracted from Dataset #2 (Henn, 2012).

		Picking time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
50	Avg (s)	20042	20215	20744	20774	21506	21912	23604	23378
	Dev (%)	0.13%	1.89%	6.60%	7.34%	14.27%	20.11%	36.03%	35.56%
	#Best	11	4	0	0	1	1	0	1
100	Avg (s)	39603	39757	39917	39919	40172	40256	40655	40633
	Dev (%)	0.13%	0.72%	2.13%	2.16%	3.37%	3.80%	6.00%	5.80%
	#Best	11	2	0	1	0	1	2	0
150	Avg (s)	58155	58115	58261	58155	58332	58344	58795	58673
	Dev (%)	0.27%	0.44%	0.77%	0.65%	1.13%	1.07%	2.19%	1.98%
	#Best	7	4	2	0	2	1	0	0
200	Avg (s)	76521	76615	76588	76693	76823	76747	77155	76872
	Dev (%)	0.26%	0.27%	0.49%	0.56%	0.85%	0.76%	1.41%	1.25%
	#Best	3	6	2	2	1	0	0	2
250	Avg (s)	95337	95154	95317	95438	95547	95454	95796	95980
	Dev (%)	0.22%	0.18%	0.49%	0.57%	0.61%	0.61%	1.21%	1.21%
	#Best	7	4	2	1	0	2	0	0

Table 8: Behavior of the picking time when varying the number of orders (congestion), for several time-window strategies (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) using the GVNS batching algorithm and the S-Shape routing strategy on the instances extracted from Dataset #1 (Albareda-Sambola et al., 2009).

		Completion time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
40	Avg (s)	20527	19578	18511	18527	18336	18328	18281	18273
	Dev (%)	14.08%	8.47%	2.12%	2.23%	1.17%	1.14%	0.75%	0.72%
	#Best	0	0	0	4	3	2	4	3
60	Avg (s)	25978	25287	24704	24728	24519	24557	24546	24508
	Dev (%)	7.60%	4.42%	1.54%	1.72%	0.83%	0.97%	0.91%	0.75%
	#Best	0	1	1	1	6	3	1	3
80	Avg (s)	32982	32390	31832	31811	31742	31669	31734	31792
	Dev (%)	5.07%	2.97%	0.90%	0.84%	0.64%	0.47%	0.67%	0.87%
	#Best	0	0	1	2	2	4	5	2
100	Avg (s)	38715	38108	37628	37684	37673	37721	37624	37747
	Dev (%)	3.73%	1.88%	0.43%	0.57%	0.53%	0.64%	0.41%	0.68%
	#Best	1	0	3	1	2	4	2	3

Table 9: Behavior of the completion time when varying the number of orders (congestion), for several time-window strategies (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) using the GVNS batching algorithm and the S-Shape routing strategy on the instances extracted from Dataset #2 (Henn, 2012).

		Completion time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
50	Avg (s)	26635	25521	24648	24633	24541	24381	24521	24288
	Dev (%)	13.24%	7.63%	2.73%	2.77%	2.06%	1.13%	1.19%	0.70%
	#Best	0	0	0	0	2	7	3	5
100	Avg (s)	42452	41848	41147	41146	41064	41089	40999	40986
	Dev (%)	5.93%	3.50%	1.16%	1.21%	0.77%	0.75%	0.65%	0.51%
	#Best	0	0	0	1	3	4	4	4
150	Avg (s)	59892	59407	59183	59081	59094	59113	59230	59114
	Dev (%)	2.67%	1.49%	0.84%	0.72%	0.66%	0.63%	0.86%	0.67%
	#Best	0	1	2	2	5	2	1	3
200	Avg (s)	78173	77922	77474	77581	77590	77490	77670	77385
	Dev (%)	1.80%	1.12%	0.45%	0.54%	0.56%	0.45%	0.56%	0.40%
	#Best	0	0	2	1	3	2	2	6
250	Avg (s)	96846	96351	96224	96362	96396	96297	96421	96647
	Dev (%)	1.17%	0.55%	0.41%	0.51%	0.41%	0.40%	0.66%	0.71%
	#Best	0	4	3	3	2	4	0	0

Table 10: Behavior of the completion time when varying the number of orders (congestion), for several time-window strategies (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) using the GVNS batching algorithm and the S-Shape routing strategy on the instances extracted from Dataset #1 (Albareda-Sambola et al., 2009).

		Maximum turnover time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
40	Avg (s)	17100	13604	11433	11847	10025	9628	10458	9297
	Dev (%)	154.41%	100.03%	54.76%	60.19%	22.83%	18.30%	23.75%	7.97%
	#Best	0	1	2	0	3	3	1	6
60	Avg (s)	22006	20437	19820	19586	18945	19005	19022	18690
	Dev (%)	38.42%	24.04%	19.27%	18.84%	15.23%	13.76%	16.55%	9.76%
	#Best	1	0	2	1	2	5	2	3
80	Avg (s)	27653	27940	26877	27079	27286	27065	26781	26917
	Dev (%)	15.17%	15.15%	11.55%	11.85%	12.51%	11.45%	10.00%	11.67%
	#Best	3	0	1	3	1	3	3	2
100	Avg (s)	34425	34161	33085	33300	33989	32864	32452	33536
	Dev (%)	15.08%	13.09%	9.29%	10.21%	12.46%	8.17%	6.46%	10.48%
	#Best	1	0	1	3	0	4	2	5

Table 11: Behavior of the maximum turnover time when varying the number of orders (congestion), for several time-window strategies (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) using the GVNS batching algorithm and the S-Shape routing strategy on the instances extracted from Dataset #2 (Henn, 2012).

		Maximum turnover time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
50	Avg (s)	23219	21542	17319	16898	15952	15490	14223	15420
	Dev (%)	361.54%	311.83%	144.50%	128.32%	33.13%	23.38%	4.51%	9.54%
	#Best	0	0	1	2	0	4	7	4
100	Avg (s)	37576	36891	34410	35450	34145	34973	34912	34827
	Dev (%)	48.68%	42.05%	18.58%	23.60%	9.18%	10.92%	15.57%	14.37%
	#Best	0	0	3	2	4	2	4	1
150	Avg (s)	56722	55610	55806	54965	55661	55196	55506	55520
	Dev (%)	8.80%	5.59%	4.83%	5.17%	4.28%	3.42%	4.62%	4.59%
	#Best	0	1	1	2	3	6	0	3
200	Avg (s)	75048	74190	73723	74862	74633	74113	74592	74075
	Dev (%)	6.22%	5.10%	3.11%	5.29%	4.78%	4.88%	4.65%	3.98%
	#Best	0	2	4	1	2	3	1	3
250	Avg (s)	93737	92925	93034	93645	93731	92555	92986	93099
	Dev (%)	4.08%	2.52%	2.19%	2.97%	2.65%	2.08%	3.03%	2.17%
	#Best	1	1	1	1	3	4	3	2

Table 12: Behavior of the maximum turnover time when varying the number of orders (congestion), for several time-window strategies (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) using the GVNS batching algorithm and the S-Shape routing strategy on the instances extracted from Dataset #1 (Albareda-Sambola et al., 2009).

630 Notice that the congestion in the system increases when, in the same time
631 horizon, the number of orders received is increased. Thus, it is observed that
632 the picking time, the completion time, and the maximum turnover time also
633 increase with the increase of congestion.

634 For the same number of orders, in all evaluated scenarios, we observe
635 significant differences between the worst and best time-window methods.
636 On the other hand, the differences between the methods are greater with
637 fewer orders in the same time horizon. As far as the objective functions
638 are concerned, it can be stated that the differences between the best and
639 worst methods are more accentuated in the case of minimizing the maximum
640 turnover time. This situation can be partially explained by the fact that it
641 looks for the minimization of a maximum value.

642 Finally, in the case of minimizing the picking time, regardless of the size
643 of the instance, the best time-window method was VTW_B3. Similarly, for
644 completion time, the most competitive methods were: VTW_M2, FTW_ZH,
645 and FTW_NW. Finally, in the case of maximum turnover time, the best
646 methods were: VTW_M1, VTW_M2, FTW_ZH, and FTW_NW.

647 **6. Conclusions and future work**

648 In this work, we have studied the influence of the time-window strategy
649 on the performance of the algorithms for the Online Order Batching Prob-
650 lem. Specifically, we have reviewed previous strategies for determining the
651 time window in the literature and selected the most prominent ones, to be
652 empirically evaluated. Also, we have proposed two new variable time-window
653 strategies.

654 Our main conclusion is that it can be stated that there is a relevant
655 influence of the time-window algorithm on the performance of the methods
656 for the OOBP when any of the following objective functions are studied:
657 picking time, completion time, or maximum turnover time. Also, we have
658 identified that the time-window algorithm should be determined depending
659 on the objective function tackled, since there is no efficient time-window
660 algorithm for all objective functions.

661 We have also demonstrated that the influence of the strategy to deter-
662 mine the time window on the objective function does not depend on the
663 batching or routing methods used. It has also been shown that the selection
664 of the best time window for each scenario does not depend on the number of
665 orders processed. However, when considering multiple pickers, the larger the
666 number of pickers, the bigger the differences among the compared methods
667 when minimizing the picking time or the turnover time. On the other hand,
668 increasing the number of pickers results in a slight decrease of the differences
669 among the compared methods, when studying the completion time.

670 The results obtained in the experimentation carried out during this study
671 indicate that, in online contexts, the minimization of the picking time can be
672 in conflict with the minimization of the completion time. This is supported
673 by the fact that, when minimizing the picking time, we look for batches
674 which can be efficiently collected. To this end, it might be necessary to
675 wait longer for the arrival of new orders, so batches can be better composed.
676 On the contrary, waiting longer for the arrival of new orders implies that
677 the total picking time is increased. This circumstance does not happen in
678 offline contexts, where the picking time and the completion time are aligned.

679 Therefore, in future works related to the OOBP, the completion time and
680 the picking time could be studied using a multiobjective approach.

681 The results obtained in this work indicate the importance of properly
682 defining the time-window strategy within the OOBP context. On average,
683 we observed that, for a single picker and an arrival time horizon of 4 hours,
684 it is possible to save up to 17 minutes in the picking time, 21 minutes in the
685 completion time, and 49 minutes in the maximum turnover time, simply by
686 choosing an appropriate time-window algorithm. Furthermore, some of the
687 above savings increase as the number of pickers also increase. For example,
688 with 5 pickers, the differences in the picking time, choosing one or the other
689 time-window method, can be up to 311 minutes (> 5 hours).

690 Among the methods for determining the time window evaluated, when
691 minimizing the picking time, it is recommended to use the VTW_B3 method,
692 while in the case of minimizing the completion time or the maximum turnover
693 time, it is recommended to use one of the following methods: VTW_M1,
694 VTW_M2, FTW_ZH, or FTW_NW.

695 Finally, we would like to highlight the importance of choosing adequately
696 the time-window method depending on the studied objective function, but
697 also on the context conditions (such as the number of pickers or the con-
698 gestion of the system). On the contrary, the selection of a time-window
699 method does not depend on the batching or routing method chosen. Given
700 the relevance of the time window, it might be interesting to explore new time-
701 window methods and their influence on other objective functions related to
702 the OOBP, not studied in this work, such as minimizing the blocking time
703 or minimizing the cost.

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807 **Appendix A. Detailed results of the influence of time-window al-**
808 **gorithms on the performance of the studied meth-**
809 **ods for different objective functions, when combined**
810 **with different routing strategies**

		Picking time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
S-Shape	Avg (s)	44017	44071	44277	44295	44564	44611	45152	45115
	Dev (%)	0.20%	0.64%	1.95%	2.03%	3.57%	4.33%	7.65%	7.57%
	#Best	76	41	9	5	5	5	3	4
Largest-Gap	Avg (s)	43730	43818	44043	44024	44284	44337	44790	44817
	Dev (%)	0.15%	0.76%	2.04%	2.06%	3.54%	4.43%	7.47%	7.46%
	#Best	93	27	4	11	4	3	2	2
Combined	Avg (s)	41390	41483	41694	41673	42519	42020	42499	42497
	Dev (%)	0.13%	0.62%	1.91%	1.87%	7.77%	4.35%	7.73%	7.74%
	#Best	90	29	6	15	2	1	1	2

Table A.13: Behavior in terms of picking time of a GVNS batching algorithm combined with different routing strategies (S-Shape, Largest-Gap, and Combined) and different time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH).

		Completion time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
S-Shape	Avg (s)	46911	46268	45706	45728	45662	45627	45670	45638
	Dev (%)	6.14%	3.56%	1.18%	1.23%	0.85%	0.73%	0.74%	0.67%
	#Best	1	6	12	15	28	32	22	29
Largest-Gap	Avg (s)	46724	46090	45568	45615	45469	45425	45406	45436
	Dev (%)	11.77%	9.10%	6.74%	8.76%	6.25%	6.13%	5.99%	6.00%
	#Best	0	0	5	52	28	19	16	26
Combined	Avg (s)	44450	43856	43324	43296	43183	43210	43163	43181
	Dev (%)	6.19%	3.58%	1.25%	1.17%	0.48%	0.67%	0.45%	0.47%
	#Best	2	3	15	14	24	41	22	30

Table A.14: Behavior in terms of completion time of a GVNS batching algorithm combined with different routing strategies (S-Shape, Largest-Gap, and Combined) and different time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH).

		Maximum turnover time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
S-Shape	Avg (s)	43054	41922	40612	40848	40485	40099	40104	40154
	Dev (%)	72.49%	57.71%	29.79%	29.60%	13.00%	10.71%	9.90%	8.28%
	#Best	6	5	16	15	18	34	23	29
Largest-Gap	Avg (s)	42970	41860	40542	40347	39960	39853	39804	39843
	Dev (%)	69.44%	48.88%	30.82%	30.60%	11.43%	11.09%	9.40%	9.68%
	#Best	6	7	16	19	25	25	24	24
Combined	Avg (s)	40633	39665	38094	38351	37557	37477	37647	37571
	Dev (%)	74.10%	54.93%	30.44%	28.95%	8.44%	10.15%	8.80%	8.71%
	#Best	9	8	14	13	23	27	24	32

Table A.15: Behavior in terms of maximum turnover time of a GVNS batching algorithm combined with different routing strategies (S-Shape, Largest-Gap, and Combined) and different time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH).

811 **Appendix B. Detailed results of the influence of time-window al-**
812 **gorithms on the performance of the studied meth-**
813 **ods for different objective functions, when combined**
814 **with different batching strategies**

		Picking time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
GVNS	Avg (s)	44017	44071	44277	44295	44564	44611	45152	45115
	Dev (%)	0.20%	0.64%	1.95%	2.03%	3.57%	4.33%	7.65%	7.57%
	#Best	76	41	9	5	5	5	3	4
ILS	Avg (s)	45820	45752	45849	45818	46123	46218	46715	46693
	Dev (%)	1.15%	1.22%	2.05%	1.96%	3.38%	4.38%	7.60%	7.58%
	#Best	41	42	27	26	12	13	4	8
FCFS	Avg (s)	49582	49595	49671	49671	49854	49932	50486	50486
	Dev (%)	0.18%	0.25%	0.68%	0.68%	1.54%	2.35%	5.49%	5.49%
	#Best	97	89	72	72	52	51	18	18

Table B.16: Behavior in terms of picking time of different batching algorithms (GVNS, ILS, and FCFS) combined with S-Shape routing strategy and different time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH).

		Completion time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
GVNS	Avg (s)	46911	46268	45706	45728	45662	45627	45670	45638
	Dev (%)	6.14%	3.56%	1.18%	1.23%	0.85%	0.73%	0.74%	0.67%
	#Best	1	6	12	15	28	32	22	29
ILS	Avg (s)	49303	48648	48055	48022	47940	47948	47968	47934
	Dev (%)	6.61%	4.08%	1.80%	1.71%	1.11%	1.08%	1.14%	1.12%
	#Best	2	5	17	14	35	20	26	30
FCFS	Avg (s)	52006	51401	50818	50818	50683	50669	50734	50734
	Dev (%)	6.06%	3.65%	1.24%	1.24%	0.59%	0.48%	0.64%	0.64%
	#Best	0	5	16	19	60	61	53	41

Table B.17: Behavior in terms of completion time of different batching algorithms (GVNS, ILS, and FCFS) combined with S-Shape routing strategy and different time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH).

		Maximum turnover time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
GVNS	Avg (s)	43054	41922	40612	40848	40485	40099	40104	40154
	Dev (%)	72.49%	57.71%	29.79%	29.60%	13.00%	10.71%	9.90%	8.28%
	#Best	6	5	16	15	18	34	23	29
ILS	Avg (s)	45231	44122	43062	42963	42150	42380	42027	42093
	Dev (%)	68.20%	45.85%	30.75%	28.84%	9.52%	9.58%	6.92%	8.12%
	#Best	14	16	18	15	18	20	24	27
FCFS	Avg (s)	46737	45408	44108	44108	43923	43972	44053	44068
	Dev (%)	65.41%	36.62%	10.82%	10.82%	8.05%	7.64%	5.71%	5.76%
	#Best	2	9	22	21	53	44	53	48

Table B.18: Behavior in terms of maximum turnover time of different batching algorithms (GVNS, ILS, and FCFS) combined with S-Shape routing strategy and different time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH).

815 Appendix C. Detailed results of the influence of time-window al-
816 gorithms on the performance of the studied methods
817 for different objective functions, when varying the
818 number of pickers

		Picking time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
1 Pickers	Avg (s)	44017	44071	44277	44295	44564	44611	45152	45115
	Dev (%)	0.20%	0.64%	1.95%	2.03%	3.57%	4.33%	7.65%	7.57%
	#Best	76	41	9	5	5	5	3	4
2 Pickers	Avg (s)	44498	44726	45272	45240	46096	47154	48824	48904
	Dev (%)	0.15%	1.26%	3.95%	3.79%	8.17%	15.47%	26.09%	26.25%
	#Best	110	23	5	5	0	0	2	0
3 Pickers	Avg (s)	44924	45275	46169	46121	47946	50646	53877	54237
	Dev (%)	0.15%	1.57%	5.36%	5.20%	12.76%	27.38%	45.96%	46.69%
	#Best	108	21	3	6	0	2	6	1
4 Pickers	Avg (s)	45174	45604	46822	46797	49531	54327	58518	59401
	Dev (%)	0.16%	1.77%	6.30%	6.10%	16.28%	39.18%	62.00%	64.06%
	#Best	117	11	5	1	0	1	9	0
5 Pickers	Avg (s)	45220	45773	47074	47043	50588	57319	62291	63918
	Dev (%)	0.15%	1.91%	6.56%	6.60%	18.53%	47.80%	74.45%	78.13%
	#Best	118	14	1	2	1	0	9	0

Table C.19: Behavior in terms of picking time of a GVNS batching algorithm, combined with S-Shape routing strategy and different time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) when varying the number of pickers.

		Completion time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
1 Picker	Avg (s)	46911	46268	45706	45728	45662	45627	45670	45638
	Dev (%)	6.14%	3.56%	1.18%	1.23%	0.85%	0.73%	0.74%	0.67%
	#Best	1	6	12	15	28	32	22	29
2 Pickers	Avg (s)	27061	26501	26095	26125	25954	25869	25840	25819
	Dev (%)	8.07%	4.98%	2.66%	2.80%	1.75%	1.24%	1.02%	0.83%
	#Best	1	1	12	10	32	28	34	34
3 Pickers	Avg (s)	21568	21169	20889	20874	20743	20549	20555	20505
	Dev (%)	7.58%	5.22%	3.52%	3.49%	2.56%	1.35%	1.17%	0.89%
	#Best	0	1	9	7	18	48	47	40
4 Pickers	Avg (s)	19334	19113	18837	18810	18625	18463	18431	18309
	Dev (%)	7.02%	5.79%	4.04%	3.90%	2.61%	1.58%	1.13%	0.54%
	#Best	2	0	4	9	23	32	49	58
5 Pickers	Avg (s)	18141	18003	17748	17721	17522	17371	17256	17181
	Dev (%)	6.72%	5.87%	4.28%	4.13%	2.76%	1.71%	0.84%	0.45%
	#Best	1	0	5	5	21	29	65	59

Table C.20: Behavior in terms of completion time of a GVNS batching algorithm, combined with S-Shape routing strategy and different time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) when varying the number of pickers.

		Maximum turnover time							
		VTW_B3	VTW_B2	VTW_B1	VTW_HE	VTW_M1	VTW_M2	FTW_ZH	FTW_NW
1 Picker	Avg (s)	43054	41922	40612	40848	40485	40099	40104	40154
	Dev (%)	72.49%	57.71%	29.79%	29.60%	13.00%	10.71%	9.90%	8.28%
	#Best	6	5	16	15	18	34	23	29
2 Pickers	Avg (s)	22547	20602	18766	18843	17451	17388	17310	17208
	Dev (%)	226.61%	154.18%	93.41%	95.33%	25.49%	19.86%	11.67%	10.54%
	#Best	5	2	10	14	19	20	37	51
3 Pickers	Avg (s)	16200	14191	12055	12199	10259	10175	10254	9910
	Dev (%)	354.37%	249.60%	138.32%	145.16%	32.72%	25.01%	15.95%	6.14%
	#Best	1	4	7	6	20	23	49	62
4 Pickers	Avg (s)	13649	11733	9390	9543	7594	7248	7415	6857
	Dev (%)	421.39%	313.18%	175.75%	177.23%	43.66%	32.19%	31.63%	4.41%
	#Best	4	1	4	5	16	16	65	75
5 Pickers	Avg (s)	12333	10667	8155	8115	6017	5798	5818	5358
	Dev (%)	479.54%	379.97%	204.43%	199.38%	51.45%	37.71%	35.19%	3.15%
	#Best	1	1	3	2	11	21	81	78

Table C.21: Behavior in terms of maximum turnover time of a GVNS batching algorithm, combined with S-Shape routing strategy and different time-window algorithms (VTW_B1, VTW_B2, VTW_B3, VTW_HE, VTW_M1, VTW_M2, FTW_NW, and FTW_ZH) when varying the number of pickers.

